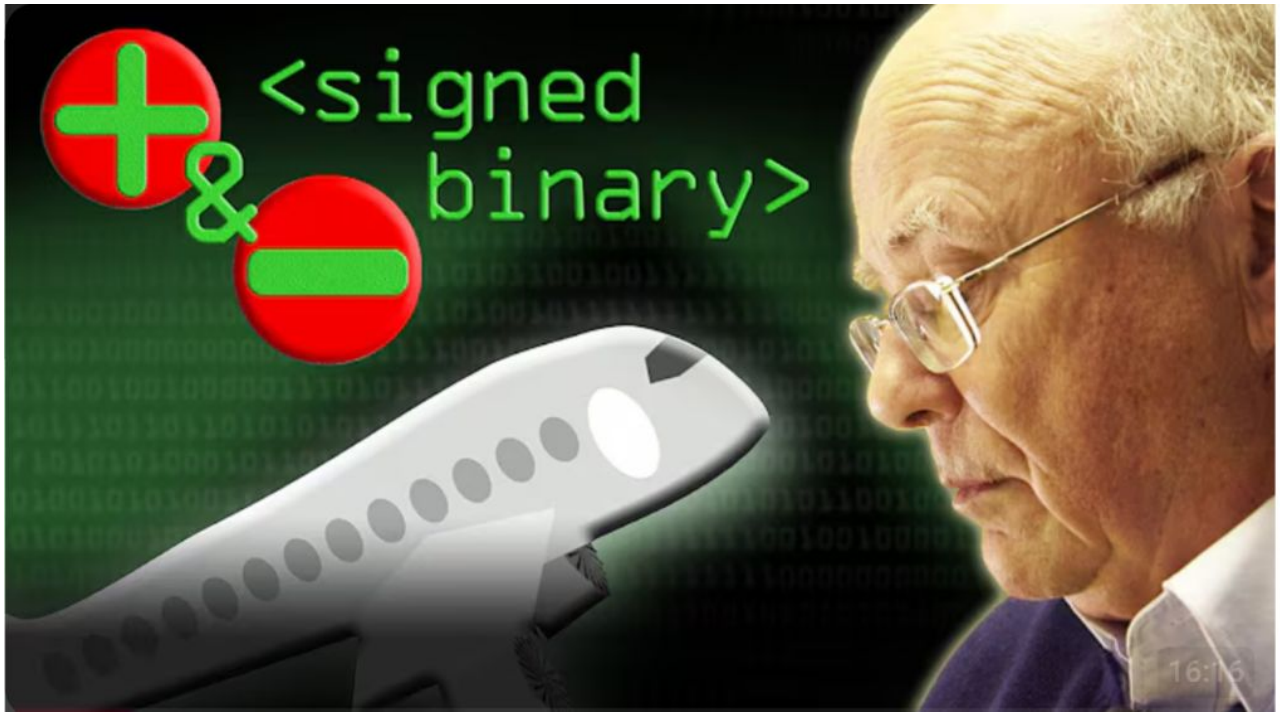
Why two's complement?

Two’s complement



Binary: Plusses & Minuses

420K views • 10 years ago

Computerphile

Negative Binary Numbers - you may have

Subtitles

5 chapters Intro | Negative

How many values can we **encode** with 3 bits?

Two’s complement

000	001	010	011	100	101	110	111
0	1	2	3	4	5	6	7
0	-1	-2	-3	-4	-5	-6	-7
0	1	2	3	-1	-2	-3	-4
0	1	2	3	-4	-3	-2	-1
-4	-2	A	55	-10	?	-5	0

Sign bit

000 **0**01 **0**10 **0**11 **1**00 **1**01 **1**10 **1**11

We look at the most significant bit (MSB) and determine the sign of the number represented with the next two bits.

001 → **+**1 **1**01 → **-**1

000 → **+**0 ?? **1**00 → **-**0

Sign bit

Not only we have meaningless negative zero, but subtraction doesn't work either:

001 + **1**10 = **1**11 1+(**-**2)=**-**3

One's complement

We apply **not** gate to each bit to represent its opposite.

001 → +1 110 → **-**1

000 → +0 ?? 111 → **-**0

000	001	010	011	100	101	110	111
+0	1	2	3	-3	-2	-1	**-0**

One’s complement

Again we waste 3 bits with unnecessary negative zero. Subtraction works though:

001 + 101 = 110 1+(**-**2)=**-**1

000	**001**	010	011	100	**101**	**110**	111
+0	**1**	2	3	-3	**-2**	**-1**	-0

Two's complement

How to make sure that both +0 and -0 are 000?

000 → 111 + 001 → ~~1~~000 → 000

Apply **not** gate and then **add 1**, remove the bit that overflows.

One's complement

We apply **not** gate to each bit to represent its opposite.

001 → +1 110 → **-**1

000 → +0 ?? 111 → **-**0

000	001	010	011	100	101	110	111
+0	1	2	3	-3	-2	-1	**-0**

Two's complement

We apply **not** gate to each bit and **add one** to represent its opposite.

001 → +1
000 → +0

111 → **-**1
000 → **-**0

000	001	010	011	100	101	110	111
0	1	2	3		-3	-2	-1

Two's complement

Subtraction works as well:

001 + 110 = 111 1+(**-**2)=**-**1

000 **001** 010 011 100 101 **110** **111**

0 **1** 2 3 -4 -3 **-2** **-1**

Two's complement

Subtraction works as well:

001 + 111 = ~~1~~000 1+(-1)=0

000 **001** 010 011 100 101 110 **111**

0 **1** 2 3 -4 -3 -2 **-1**

Exercise

See if **3 – 7** indeed returns **-4** in binary.